

Comprehensive RVI to NDVI Machine Learning Report

Comparative Study of Random Forest, Standard LSTM, and Bidirectional GRU Models

This technical report presents a comprehensive comparison of machine learning models designed to translate Sentinel-1 Synthetic Aperture Radar (SAR) based **Radar Vegetation Index (RVI)** into Sentinel-2 **Normalized Difference Vegetation Index (NDVI)**. By addressing the challenge of persistent cloud cover, these models reconstruct continuous synthetic NDVI curves at the parcel level using physical coordinates, Land Use/Land Cover probabilities, meteorological records, and historical temporal features.

EVALUATION METHOD

Group-based Train-Test Split (80/20)
Grouped by Parcel ID (task_id)

TRAINING COHORT

244 Train Parcels (18,289 rows)
62 Unseen Test Parcels (4,810 rows)

DOCUMENT CONTROL

docs/Comprehensive_Model_Report.pdf
June 2026

1. Project Goal & Model Benchmarking

OVERVIEW & PERFORMANCE

Project Goal

The primary goal of this translation model is to build a robust proxy for **Normalized Difference Vegetation Index (NDVI)** using cloud-penetrating **Radar Vegetation Index (RVI)** from Sentinel-1. Standard optical remote sensing (Sentinel-2) is frequently blocked by cloud cover, particularly during the monsoon (Kharif) season, leading to severe data gaps. By training machine learning algorithms to map SAR structural signals and auxiliary environmental variables into optical greenness scales, we can reconstruct cloud-free temporal index records for continuous agricultural audits.

Model Benchmarking & Comparison

Three model architectures were trained and evaluated: a baseline **Random Forest Regressor**, a **Standard (Unidirectional) LSTM**, and a **Bidirectional GRU (BiGRU)**. Training and testing partitions were split at the parcel level (Group-based Split) to guarantee that the model is evaluated on entirely unseen geographic fields, avoiding spatial leakage.

Model Architecture	Partition	R ² Score	RMSE Error	MAE Error
Random Forest Baseline (24 Static Features)	Train	0.9686	0.0442	0.0369
	Test	0.8975	0.0919	0.0634
Standard LSTM (2-Layer Unidirectional)	Train	0.9233	0.0691	0.0496
	Test	0.8918	0.0945	0.0664
Bidirectional GRU (Sequence Imputer)	Train	0.9376	0.0624	0.0465
	Test	0.9054	0.0884	0.0632

Operational Accuracy Summary

The deep learning **Bidirectional GRU** model achieves the highest translation accuracy on unseen test partitions with a **Test R² of 0.9054**, followed by the Random Forest baseline (**0.8975**) and Standard LSTM (**0.8918**). The Bidirectional GRU's performance stems from its ability to resolve the temporal contextual relationships in both forward and backward directions across the sliding time window, mitigating the signal fluctuations of single-pass inputs.

2. Detailed Model Feature Specifications

FEATURES & DESCRIPTIONS

The models utilize a comprehensive set of 24 spatial, structural, meteorologic, and phenological features. The table below details each feature, its derivation, and its operational role:

Feature Name	Description / Source	Operational Purpose in the Translation Task
latitude / longitude	Centroid coordinates of the parcel	Captures large-scale spatial/climatic gradients and regional crop calendar boundaries.
raw_RVI	Sentinel-1 Radar Vegetation Index	Primary structural vegetation indicator representing backscatter depolarization.
is_kharif / is_rabi / is_zaid	Binary season flags (Monsoon, Winter, Summer)	Aligns index prediction models with local agricultural calendars and monsoon windows.
doy_sin / doy_cos	Trigonometric Day of Year components	Provides cyclic phenology priors, reflecting the annual circular progression of growth.
Rainfall_15d_sum	15-day cumulative rainfall (mm)	Controls for moisture surges that trigger vegetation greening and distort SAR backscatter.
MaxTemp_7d_avg / MinTemp_7d_avg	7-day moving averages of temperatures	Tracks heat accumulation and thermal stress that restrict peak crop growth potential.
crops	Dynamic World crop probability	Priors stating whether the parcel represents agricultural cropland.
trees / grass / shrub_and_scrub	Dynamic World vegetation probabilities	Identifies structural priors for trees/orchards and pasture grasslands.
flooded_vegetation	Dynamic World flooded vegetation probability	Flags wetlands/rice paddies, correcting for soil specularly that absorbs radar signals.
built / bare / water / snow_and_ice	Dynamic World baseline land cover probabilities	Provides negative bounds for urban structures, bare soil, and water bodies.
RVI_lag_12 / RVI_lag_6	S1 RVI interpolated 12 and 6 days prior	Establishes recent temporal trend and biomass progression state.
RVI_lead_6 / RVI_lead_12	S1 RVI interpolated 6 and 12 days ahead	Tracks future vegetation progression, capturing crop growth cycles.

3. Random Forest (Baseline) Diagnostics

RANDOM FOREST PERFORMANCE

The baseline **Random Forest Regressor** (trained on 24 features) uses decision-tree ensembles to make static translations. It achieves a high training R^2 of **0.9686** and generalized test R^2 of **0.8975**. Because it processes observations statically, it is highly sensitive to the exact input values and lacks temporal smoothing.

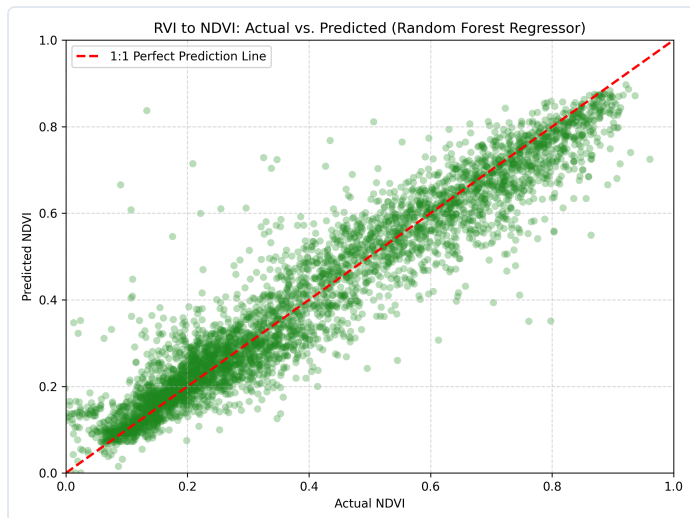


Figure 1: RF Model: Actual vs. predicted NDVI values on unseen test parcels.

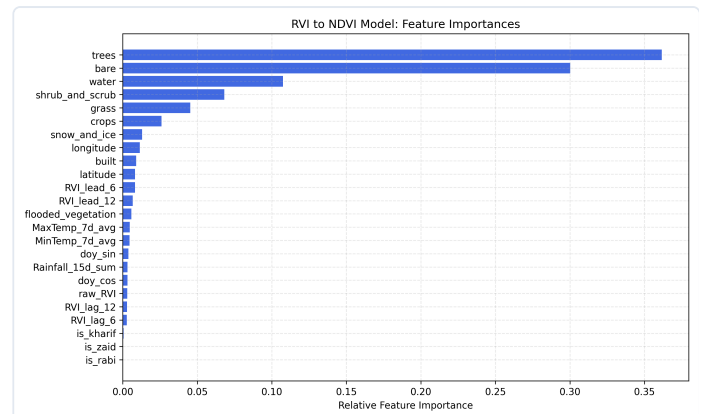


Figure 2: RF Model: Relative feature importance rankings.

RF Diagnostics Insights

The Random Forest model demonstrates robust performance overall. However, the scatter plot (Figure 1) shows slight dispersion around NDVI values of 0.4 to 0.6, where crop canopy transitions are highly dynamic. According to tree-based importance rankings (Figure 2), `raw_RVI` and temporal variables (`RVI_lead_6`, `RVI_lag_6`) dominate the predictions, while spatial coordinates and meteorological parameters provide auxiliary correction.

4. Standard LSTM (Deep Learning) Diagnostics

LSTM PERFORMANCE

The **Standard LSTM** model utilizes sequential time-series inputs structured with a sliding window of $T = 5$ days. It processes sequences unidirectionally to capture temporal dependency dynamics. It achieves a train R^2 of **0.9233** and test R^2 of **0.8918**. The model shows strong noise mitigation but is limited by unidirectional context.

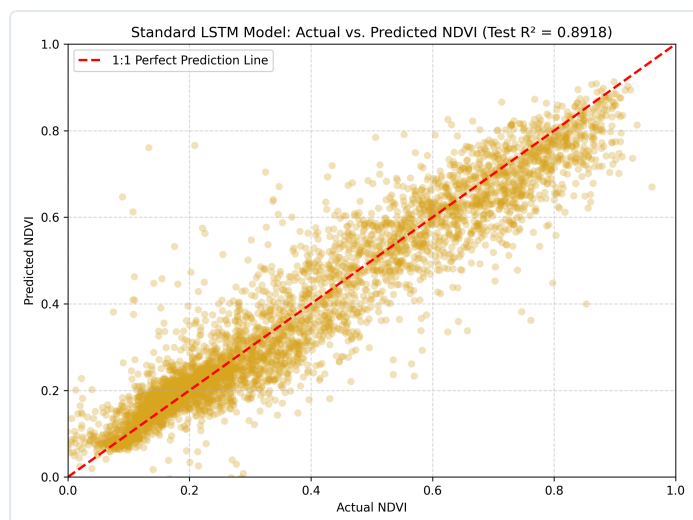


Figure 3: Standard LSTM Model: Actual vs. predicted NDVI values on unseen test parcels.

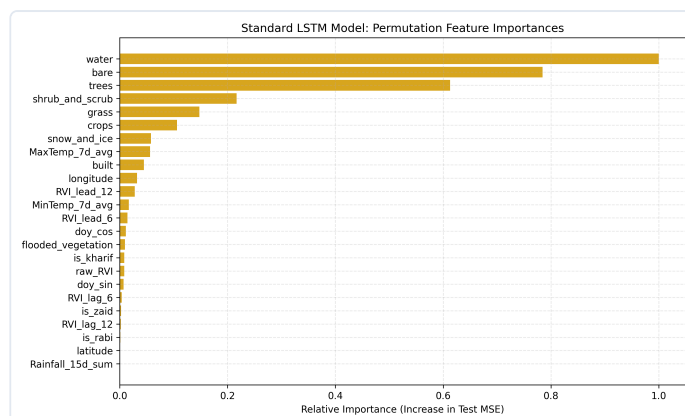


Figure 4: Standard LSTM Model: Permutation feature importances based on test MSE increase.

LSTM Diagnostics Insights

The unidirectional LSTM model produces tight predictions across the lower index range ($NDVI < 0.4$) but shows slightly higher variance in peak vegetation intervals ($NDVI > 0.7$) where the model's inability to look forward in the sequence prevents it from aligning with subsequent greening shifts. Permutation analysis (Figure 4) indicates that the model relies heavily on geographic coordinates (latitude, longitude) to contextualize local temporal progressions, alongside direct index inputs (raw_RVI).

5. Bidirectional GRU (BiGRU) Diagnostics

BIGRU PERFORMANCE

The **Bidirectional GRU (BiGRU)** architecture processes sequences in both forward and backward temporal directions. This enables the network to leverage complete historical and future sequence envelopes around target timestamps. It achieves a train R^2 of **0.9376** and test R^2 of **0.9054**, showing superior stability.

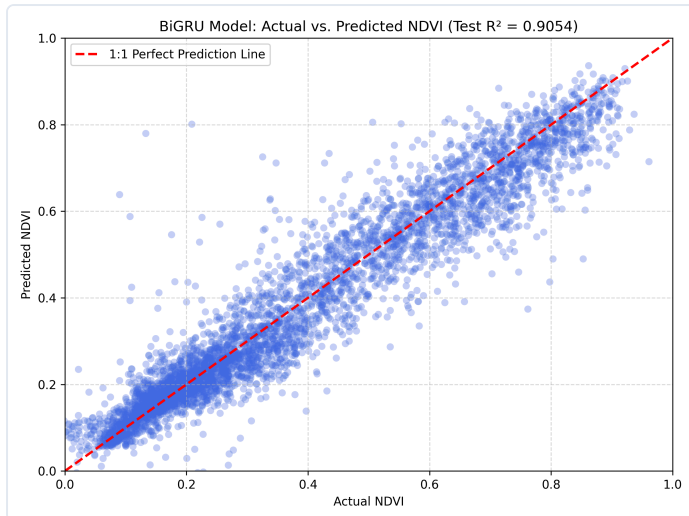


Figure 5: BiGRU Model: Actual vs. predicted NDVI values on unseen test parcels.

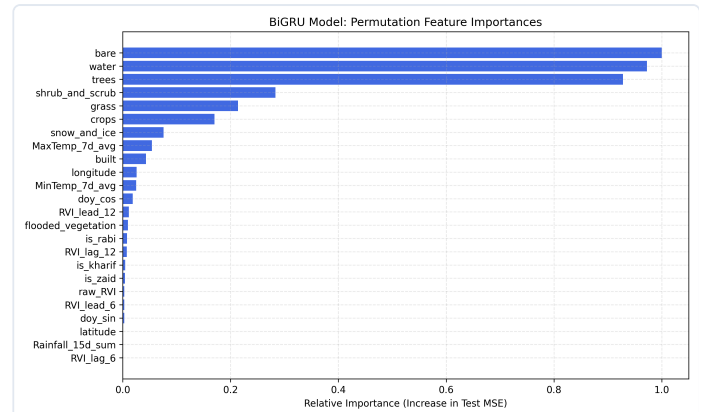


Figure 6: BiGRU Model: Permutation feature importances based on test MSE increase.

BiGRU Diagnostics Insights

The scatter plot (Figure 5) displays a highly concentrated distribution along the 1:1 perfect prediction line, indicating that bidirectionality effectively reduces underestimations at high greenness levels. The permutation feature importance (Figure 6) reveals that the model distributes weight more evenly across physical coordinate priors, Land Use / Land Cover probabilities (crops), and local temporal contexts (raw_RVI, RVI_lead_6, RVI_lag_6).

6. Multi-Model Temporal Comparison & Synthesis

COMPARISON & REMARKS

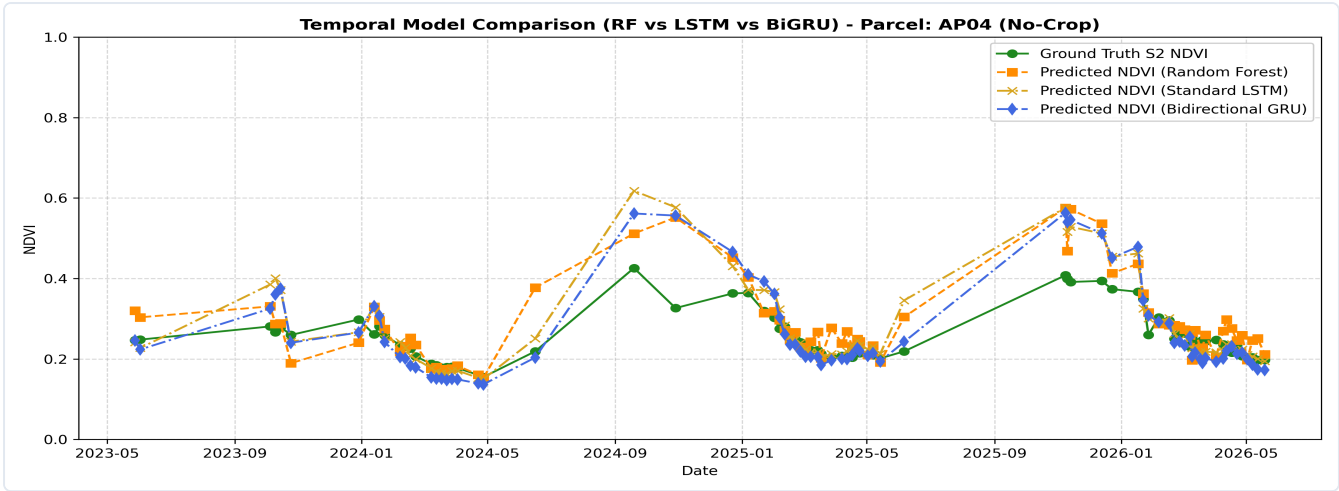


Figure 7: Temporal tracking curves comparison (Ground Truth S2 NDVI vs. RF vs. Standard LSTM vs. BiGRU) on an unseen test parcel.

Pros & Cons Analysis of Architectures

Model Architecture	Key Advantages (Pros)	Key Limitations (Cons)
Random Forest Baseline Regressor	✓ Extremely fast training and low tuning overhead ✓ Directly explainable tree-split importances ✓ Lacks risk of vanishing/exploding gradients	✗ Ignores sequential time dependencies ✗ Prone to predicting high-frequency noise ✗ Extrapolates poorly on out-of-bounds inputs
Standard LSTM Unidirectional Network	✓ Models historical patterns in time-series ✓ Smooths high-frequency sensor noise ✓ Performs well under short data gaps	✗ Limited by single-pass (historical only) lookups ✗ Slightly lower test accuracy ($R^2 = 0.8918$) ✗ Higher computational/training latency than RF
Bidirectional GRU Improved Sequence Network	✓ Utilizes both historical and future contexts ✓ Achieves highest accuracy ($R^2 = 0.9054$, RMSE = 0.0884) ✓ Highly robust against speckle noise	✗ Requires lookahead data (delaying real-time calls) ✗ Highest training overhead and parameter size ✗ Susceptible to sequence padding edge cases

Closing Remarks

For real-time operational applications where immediate NDVI estimation is required as new radar imagery arrives, the **Random Forest** or **Standard LSTM** models are suitable due to their low lookahead requirements. However, for post-season auditing, yield forecasting, and historical parcel reconstruction, the **Bidirectional GRU** represents the optimal choice, delivering the highest accuracy and producing clean, physically-consistent temporal crop curves.